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Political inclusion and educational investment: Estimates from a national policy experiment in India

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ABSTRACT

I examine whether the "role model effect" of female politicians on adolescent girls' school enrollment comes only from female representation in the lowest level of local government. To do this, I exploit variation in the implementation of seat quotas for women applied to a three-tiered local governance structure in India. Enrollment effects were substantially larger than those previously estimated for exposure to female leaders in the lowest tier of local government. The policy response is larger among girls in poorer households and those with less-educated women in the household, and was commensurate with reductions in idle time and employment in household enterprises. There is no evidence of additional school infrastructure, nor a reallocation of schooling resources. Effect magnitudes imply that more than two thirds of the net effect of the policy comes from female politicians serving in reserved positions other than the village council chairperson seat.

1. Introduction

Existing research has convincingly found substantial policy and welfare effects that result from the imposition of seat quotas for women in politics (Chattopadhyay and Duflo, 2004; Iyer et al., 2012; Kalsi, 2015, among others). Other work has demonstrated that exposure to female leaders in government changes the aspirations of and beliefs about women (Beaman et al., 2009a), and that exposure to female leaders can result in changes in typically hard-to-influence behaviors such as human capital investment (Beaman et al., 2012) and trust in institutions (Iyer et al., 2012) – implying longer-lasting effects on young women than those expected via changed policy actions alone.

While pioneering studies have largely focused on the policy and welfare effects of exposure to female leaders at the village level, the quota policy was, in fact, concurrently applied to the three tiers of local government. In particular, whether the much-celebrated "role model effects" on girls' school enrollment similarly occur as a result of exposure to quota-mandated female leaders in higher levels of government remains an open question. This paper aims to address this gap by asking whether the level of government at which a political seat quota policy is implemented matters in inspiring young women to invest in their education. Knowing whether this is the case is particularly important for voters, governments, and development practitioners who face tradeoffs in the choice of whether and where to implement quotas.

In this paper, I investigate whether girls' educational enrollment responds to the presence of quota-mandated female politicians in Indian local government. To do this, I exploit differential timing of the implementation of the quotas across states to estimate a border fixed-effects identification strategy that identifies the effects of exposure to the quotas regime based on cumulative differences in policy exposure across states arising from the staggered state-level implementation of the quota policy (Iyer et al., 2012; Ghani et al., 2014; Kalsi, 2015).

I find that the effect of policy exposure on enrollment rates of young women is substantial: an additional year of exposure to the quota regime increases girls' school enrollment rate by 1.6 percentage points, with a negligible effect on young men. Findings regarding children's labor force participation and household responsibilities show that enrollment increases replaced idle time and non-labor force household duties. I use rich information on household characteristics to show that effects were strongest among poor families and those with uneducated adult women. Further analyses then explore the channels through which exposure to the quota policy changed enrollment. Using detailed administrative data on school infrastructure, I show that there is no evidence of an increase in either the extensive or intensive margins

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of schooling infrastructure resulting from longer exposure to seat quotas, nor in within-district inequality in educational infrastructure which would be indicative of within-district resource reallocation. I also find no effects on the educational attainment among older cohorts who had completed schooling by the time the policy was introduced, nor evidence of selective migration. The estimated per-year effect is large and contributed to a substantial closing of the gender enrollment gap over the period.

I also discuss whether the effects are due only to council chairwomen at the lowest (village) level of local government, or whether other female politicians put into office because of the quota policy may have had an additional effect on young women's enrollment decisions. While there is no direct empirical test of this available, per-year exposure to female leaders in office (across all levels of local government) are more than 50 percent larger then the previously estimated magnitude of the effect of an additional year of exposure to a village council chairwoman. This also largely rules out the possibility that enrollment effects were derived from exposure to village chairwomen only, as this would imply an implausibly large effects on a per-year-exposed basis. This suggests that a substantial portion (more than two thirds) of the net effect estimated in this paper was due either to other female (non-chair) council members and/or female chairwomen in higher levels of local government.

The next section describes essential aspects of the reservations policy and discusses related literature. I then discuss the empirical approach and data, and present results. This is followed by an investigation into heterogeneous effects, displaced activities, and causal channels. The final section concludes with a discussion of how results complement the findings of earlier literature and provides avenues for future research.

2. Background

Passed in 1992, India's 73rd Constitutional Amendment formalized national support for a decentralized governance structure known as the panchayat. Importantly for this study, the 73rd Amendment instituted a three-tiered system of local government councils at the village, town, and district levels across rural areas of the country that required one-third of seats at each governance level within local government to be filled by women (hereafter, the "one-third rule"). The one-third rule applied both to the council bodies (which had a uniform one-third quota for ward members) as well as single-seat leadership positions within councils (i.e., council chairperson seats, the reservation of which rotated across constituencies at different times). The Amendment also reserved a proportionate seat share for representatives from historically marginalized social groups (Scheduled Castes and Scheduled Tribes, hereafter noted by "SC/ST"). For further detail on this history and implementation of the 73rd Amendment, I refer the reader to previous studies that examine this policy and its effects, including Beaman et al. (2012), Iyer et al. (2012), Ghani et al. (2014), and Kalsi (2015), among others.

The 73rd Amendment stipulated that states had the responsibility to adjust or amend local elections to comply with the provisions of the Amendments, and nearly all states amended existing laws or passed new laws to be compliant within one year. However, there is considerable variation in the timing of these first elections across states. As first pointed out by Iyer et al. (2012), this timing is plausibly exogenous due to state authorities waiting for the term of existing governing bodies to expire. That is, states waited for the term of office of incumbent local officials to expire before conducting fresh elections in compliance with the provisions of the reform. Although comprehensive data do not exist for all states, available evidence confirms there was only minimal involvement of women in local elected bodies prior to this reform (Mathew, 2000). The policy thus represented a substantial and sharp change in the participation of women in the public sphere in rural India.

In an influential work, Beaman et al. (2012) use the village-level random assignment of leadership (chairperson) positions to women among a sample of households in 495 villages in West Bengal to find that chairwomen substantially reduced the adolescent educational enrollment gap between boys and girls. This reduction was on the order of a one percentage-point decrease in the enrollment gap per year of exposure to a chairwoman in the village council, and was attributed to the fact that the gender gap in aspirations closed among both parents and adolescents.

While the work above isolates a credible estimate of the effect of female leaders at the village level, the quota system operated in a three-tiered government structure – and an open question remains as to whether quotas operating at the higher (town and district) levels of local government, or among council (ward) members at any of the levels, have similar effects. One hypothesis could be, for example, that the relative power of a position could have a similar or larger effect than that of the village leader; alternatively, female leaders who are more proximate to young adults – either on a contact basis, or in terms of socioeconomic similarity – may be the most effective in increasing enrollment through role model effects.[1]

This paper thus seeks to shed light on whether role model effects of female leaders are derived from those serving in positions other than the village chairperson seat. To do this, I use the staggered manner in which the quota policy was implemented across states to estimate the effect an additional year of exposure to the quota regime had on children's enrollment. Given the one-third rule, this approach yields an estimate the intent-to-treat (ITT) effect of exposure to female politicians, rather than the average treatment effect (ATE) effect of exposure to village chairwomen estimated in previous work. Because the quota policy was applied to three levels of local government, however, each year of policy exposure yields, in expectation, one year of exposure to a female chairperson in local government. This allows me to directly compare coefficient magnitudes estimated below with previous estimates to infer whether the combined effects of exposure across levels is equal to, or different from, previous estimates of the village leadership ATE.

3. Empirical approach

A first-best approach to estimating the role model effects of female leadership would map individual students' behavior to variation in exposure to female leaders at each of the levels of government in which quotas were operating. Government records were not kept in such a way to comprehensively observe the village and town reservations status historically, so for empirical identification I instead exploit differential timing of the policy implementation across states. This information is then combined with household survey data that provides individuals' district of residence, information on school enrollment, and labor force activities for a nationally-representative cross-section of the population.

Taking advantage of the staggered rollout of the quota policy across states, I use differences in the cumulative years of exposure to the policy across neighboring districts lying on either side of state borders to identify the effect of exposure to the quota policy on children's enrollment outcomes.[2]

Districts typically have multiple cross-border neighbors, and the primary estimates come from a sample of cross-border pairs constructed

[1] Nguyen (2008), for example, finds that role model identity matters in the effect of information on updating perceived returns to education: role models of similar backgrounds to students have a larger impact on outcomes than role models of dissimilar backgrounds.

[2] Similar identification strategies have been implemented in many contexts; see for example, Michalopoulos and Papaioannou (2014), Ghani et al. (2014), Jofre-Monseny (2012), Duranton et al. (2011), Naidu (2012), Dube et al. (2010), and Holmes (1998).

Table 1
Mean enrollment rates in rural areas, by sex, age group, and survey year.

Age group	1987	1993	1999	2004	2007	Change
Panel A: Male						
5–10	0.35	0.69	0.73	0.82	0.84	0.49
11–15	0.63	0.71	0.74	0.82	0.85	0.22
16–18	0.35	0.39	0.41	0.44	0.49	0.14
Panel B: Female						
5–10	0.28	0.57	0.64	0.78	0.82	0.54
11–15	0.37	0.50	0.60	0.70	0.77	0.39
16–18	0.12	0.18	0.24	0.3	0.38	0.26
Panel C: Enrollment gap						
5–10	−0.08	−0.13	−0.09	−0.04	−0.02	0.05
11–15	−0.26	−0.21	−0.14	−0.12	−0.08	0.18
16–18	−0.23	−0.22	−0.16	−0.14	−0.11	0.12

Note: Author's calculations using data from the National Sample Survey Organisation, Schedule 10: Employment/Unemployment (various rounds). Enrollment rates are calculated as the mean across individuals of an indicator for usual principal activity reported as "attending an educational institution" for the age groups specified. Population estimates are constructed by weighting by the inverse sampling probability (sample weights) provided with the data.

to ensure that each district appears in the sample only once, as part of one pair. When constructing a pair for a given district from among multiple neighbors, the assignment to a pair is made based on the pair formation that yields the minimum value in the sum of squared differences among unit-standardized measures of observables from the 1991 Population Census data (the rural population share, the SC/ST population share, and the rural female literacy rate). The two districts forming the "matched pair" are then included in the sample and removed from the pool of possible matches for other districts. Each unique pair is thus comprised of the two most similar districts among possible adjacency combinations. The empirical advantage of this process is that a given district can appear only once in the estimation sample as part of a unique pair, such that there is no mechanically induced error term correlation resulting from the inclusion of the same district multiple times – where bootstrap adjustments for standard errors in the general context of matching with replacement have been shown to be invalid (Abadie and Imbens, 2008). In the current case, whether non-duplicative pair sets are constructed based on matching observables or are randomly chosen among multiple neighbors does not make a substantive difference in the pattern or magnitudes of the estimated policy effects, or conclusions therefrom.

Once border pairs are identified, the estimating equation takes the following form:

$$enrolled_{spji} = \delta_0 + \delta_1 exposure_s + \left(\mu_p + \epsilon_{spji}\right) \quad (1)$$

Where i indexes individuals residing in district j, which is simultaneously part of district pair p and contained in state s. The cumulative years of exposure to the policy in state s is captured by $exposure_s$, and the error term component μ_p contains unobservables common to the district pair, while the district-specific disturbance term ϵ_{spji} captures factors determining enrollment that may be distinct across municipalities within the area but are exogenous to the policy exposure.

The identifying assumption in equation (1) is that $exposure_s$ is uncorrelated with the error term, conditional on a vector of fixed effects μ_p. This approach relies on the argument that unobserved factors that determine enrollment and may be correlated with policy exposure (such as tradition, culture, social norms, and economic structure) are similar in districts that straddle state borders, and that the inclusion of a vector of pair fixed effects (μ_p) will achieve conditional independence. Violations of this assumption are likely to come in the form of district-specific unobservables that are not purged by the pair fixed effects. To address this concern, I estimate specifications that include parametric controls that proxy for various unobservables of concern. In all specifications, I also include a control for the minority population share interacted with the cumulative years of exposure to the policy and a SC/ST indicator to control for the effect of concurrent proportionate ethnic quotas in local government. The inclusion of any single or set of controls only makes estimates larger and/or more precise, allaying concern that accounting for these unobservables would meaningfully alter the conclusions of the analysis.

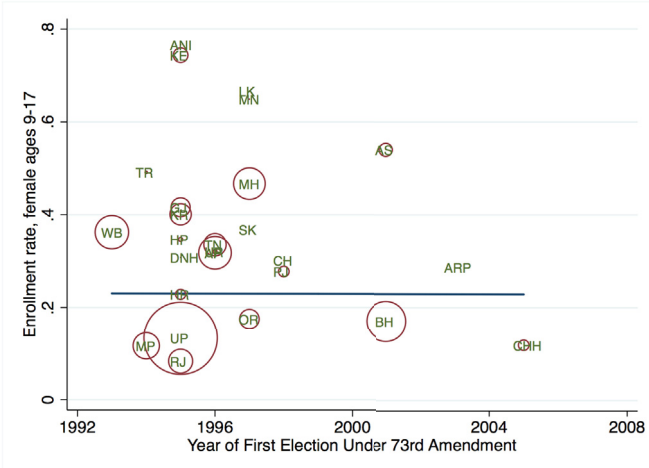
4. Data and design validity

Data on the timing of exposure to seat quotas comes from several publications documenting the implementation and progress of the 73rd Constitutional Amendment (Mathew (1995), Mathew (2000), GOI (2008), Iyer et al. (2012), Kalsi (2015)). India's National Sample Survey Organisation's Socioeconomic Survey (Schedule 10.2: Employment and Unemployment) provides individual-level data on children's and young adults' current enrollment status. These data are based on a survey of a representative sample of households across all Indian states and union territories, with the household sampling frame drawn from the most recent population census and stratified within the rural and urban areas in each district.

The Socioeconomic Survey is akin to many household labor force surveys administered worldwide. Respondent households provide individual-level details on demographics, employment, income and consumption particulars for all household members. The analysis exploits variation in the "usual principal activity" field, which, among other activities, indicates whether the individual was currently "attending [an] educational institution." Beyond restricting to the sample of school-age children in rural areas of the country, minimal data cleaning procedures were required aside from geographic definitions being made consistent over time to account for changes in administrative boundaries and the bifurcation of three states in 2001.[3]

Table 1 shows the trend in enrollment rates for different age groups over the period of study based on NSS survey data rounds from 1987 to 2007. There has been substantial progress in enrollment rates in all age cohorts between 1987 and 2007, with boys aged 5–10 and 11–15 improving their enrollment rates by 49 and 22 percentage points, respectively, while girls saw a larger increase of 54 and 39 percentage points over these two decades. In all age groups, girls lag behind boys

[3] I use the 2007-8 survey year as the relevant outcome period because after 2007, several of the later-adopting states increased seat quotas for women in local government to a minimum of 50 percent. I thus restrict the analysis to the 2007–08 wave due to incomparability of the policy measure and the exogeneity of its implementation across states after this period.



Notes: Figure depicts pre-policy enrollment rate for girls 9-17 years old by state relative to states' eventual policy implementation year. Source: Author's calculations based on GOI (2008) and data from NSS Schedule 10, 1987-88. Bubble sizes represent underlying population estimates. Coefficient: -0.00016 (S.E. = 0.012)

Fig. 1. Pre-policy female enrollment rates by state, 1987.

in initial and ending enrollment levels, although the increases in enrollment rates were greater among girls (shown in Panel C), as the enrollment gap closed by 18 percentage points in the adolescent age group (11–15 years old).[4]

In order to use state-level implementation as exogenous identifying variation, I look for evidence of whether the implementation timing across states was related to pre-period enrollment rate levels. Fig. 1 shows the initial state-level of the rural female enrollment rate by state in 1987, plotted against the year in which states implemented the quota policy. The line is virtually flat, supporting the claim that the reform implementation was unrelated to the level of pre-period female enrollment rates.[5]

I then investigate whether there is empirical support for the parallel trends assumption by directly estimating the enrollment status of individuals using data from the earliest available survey year (1983–84) through the year of the reform passage (1993–94) with a linear trend term and an interaction of this trend with the state-level year of implementation. State fixed effects absorb the main effect of differential adoption years by state, and state-clustered standard errors correct for serial correlation within states. In this specification, a negative coefficient on the interaction term would indicate that states implementing the policy earlier also had more quickly rising enrollment rates prior to the policy change. In Table 2, I find no systematic evidence of the trend in enrollment rates for the overall sample of school-age children (Panel A), nor any of the subsamples by different age groups (Panels B–D).

Finally, I show that the exposure measure is unrelated to other observable characteristics in the cross-section by regressing the exposure measure on the middle-school completion rates of women, the SC/ST population share, the sex ratio, and the area's urbanization level based on the 1991 Population Census. I also include measures of exposure to a large-scale national education program (The District Primary Education Program, or DPEP) and state progressive-party rule from

Table 2
Testing for differential trends in pre-policy enrollment rates.

	(1) Enrolled	(2) Enrolled
Panel A: Full Sample		
Year * State Implementing Year	0.00043	0.00056
	(0.00056)	(0.00050)
Year	0.01828***	0.01424***
	(0.00207)	(0.00191)
Observations	179,888	204,687
R^2	0.116	0.050
Panel B: 5–10 yrs. old		
Year * State Implementing Year	0.00031	0.00056
	(0.00083)	(0.00094)
Year	0.02287***	0.01987***
	(0.00312)	(0.00397)
Observations	89,781	100,075
R^2	0.152	0.101
Panel C: 11–15 yrs. old		
Year * State Implementing Year	0.00071	0.00014
	(0.00046)	(0.00021)
Year	0.01773***	0.01344***
	(0.00212)	(0.00127)
Observations	56,964	67,673
R^2	0.139	0.049
Panel D: 16–18 yrs. old		
Year * State Implementing Year	0.00056	0.00091
	(0.00059)	(0.00065)
Year	0.00697***	0.00539
	(0.00242)	(0.00375)
Observations	33,143	36,939
R^2	0.111	0.033

Note: Table reports a test of differential pre-period trends estimated via a linear probability model regressing individual-level enrollment status of school-age children in rural areas across three pre-intervention survey waves on a linear time trend and the time trend interacted with the state-level policy implementation year. Heteroskedasticity-consistent robust standard errors clustered by state reported in parentheses. All specifications include an unreported constant term and vectors of state fixed effects. Estimations are weighted by the provided sampling weight. Significance levels are indicated by $^{*}<.1$, $^{**}<.05$, $^{***}<.01$.

1992 to 2007. Using a specification analogous to equation (1), Table 3 presents the coefficients on each of these measures in explaining eventual exposure conditional on pair fixed effects. These are estimated separately in columns 1–7, and then jointly in column 8. None of the measures individually predict exposure levels (implementation timing) and the joint hypothesis test of the coefficients in column 8 yields an F-statistic that fails to reject the joint null hypothesis (p-value: 0.34), providing further confirmation that the border districts used do not exhibit any observable characteristics related to eventual exposure.

Table 4 provides summary statistics across individual-level records in the covariate-matched sample, separated by age groups. Appendix Table 1 provides evidence that border districts are similar to interior districts in demographics and enrollment rates both prior to the policy change in 1987 and as of 2007.

4.1. Results

Table 5 contains the results from the estimation of equation (1) for the sample of 11–15 year-olds, with all estimations weighted by the provided sample weight and standard errors clustered by state. Estimates of effects on the female subsample are in Panel A, the male subsample in Panel B, and the difference (estimated through a fully-interacted specification on the full sample) is in Panel C. Coefficient magnitudes are interpreted as the fractional increase in the enrollment rate per additional year of policy exposure. The base specification (Column 1) suggests that an additional year of exposure to the policy increases

[4] Azam and Kingdon (2013) characterize the extent and patterns of sex bias in education in India. Notably, they find that pro-male gender bias exists in the primary school age group and increases with age. The authors also find that the extent of pro-male gender bias in educational expenditure is substantially greater in rural than in urban areas.

[5] There is similarly no relationship between pre-period enrollment rates and eventual implementation year when using NSS data from the subsequent round in 1993.

Table 3
Testing for relationship between area characteristics and exposure.

	SC/ST population share (1)	Sex ratio (2)	Rural population share (3)	Female middle school completion rate (4)	DPEP Exposure (5)	1987 enrollment rate (6)	Years of Congress party (7)	All (8)
SC/ST population share	−2.201							0.011
	(4.944)							(3.589)
Sex ratio		−11.241						−9.628
		(8.936)						(6.802)
Rural population share			−4.199					−3.055
			(4.626)					(3.877)
Female middle school completion rate				3.741				5.260
				(6.749)				(6.197)
DPEP Phase 1 district					−0.074			0.457
					(0.904)			(0.701)
DPEP expansion district					−0.832			−0.213
					(0.751)			(0.556)
1987 district enrollment rate						0.905		0.691
						(0.870)		(0.446)
Years of Congress-party rule							−0.091	−0.076
							(0.166)	(0.133)
N	40,699	40,699	40,699	40,699	40,699	40,699	40,699	40,699
R^2	0.01	0.05	0.03	0.01	0.02	0.06	0.02	0.15
p-value, testing joint null					0.42			0.34

Note: Table presents a falsification test predicting eventual policy exposure with pre-policy area characteristics. Estimates via OLS. Sample comprised of 178 covariate-matched border districts as described in text. All specifications include an unreported constant term and district pair fixed effects. Significance levels are indicated by $^{*}< .1$, $^{**}< .05$, $^{***}< .01$.

Table 4
Summary statistics cross-border estimation sample.

Variable	Mean	Std. Dev.	Min.	Max.	N
Panel A: 5–10 yrs. old					
Enrolled	0.827	0.378	0	1	18,294
Years policy exposure	10.585	2.923	2	14	18,294
Female	0.467	0.499	0	1	18,294
Age (years)	7.544	1.733	5	10	18,294
SC/ST	0.347	0.476	0	1	18,294
Sample weight	2375.4	2228.7	9.7	24935.9	18,294
Panel B: 11–15 yrs. old					
Enrolled	0.789	0.408	0	1	13,789
Years policy exposure	10.692	2.844	2	14	13,789
Female	0.459	0.498	0	1	13,789
Age (years)	13.090	1.366	11	15	13,789
SC/ST	0.337	0.473	0	1	13,789
Sample weight	2339.3	2235.1	11.1	25601.8	13,789
Panel C: 16–18 yrs. old					
Enrolled	0.421	0.494	0	1	8616
Years policy exposure	10.730	2.815	2	14	8616
Female	0.452	0.498	0	1	8616
Age (years)	17.096	0.862	16	18	8616
SC/ST	0.340	0.474	0	1	8616
Sample weight	2177.3	2193.3	11.9	19296.4	8616

Note: Table shows summary statistics for individuals living in the estimation sample of matched districts lying on state borders.

the enrollment rate of young women by 1.7 percentage points. The pre-policy enrollment rate for this sample was less than 50 percent (Table 1), and mean exposure to the policy was 10.6 years, implying the policy played a substantial role in the overall increase in the educational enrollment of young women over this period. Panel B contains estimates of the effect of policy exposure among young men, showing minimal commensurate effects among the male peers in this age group. The difference between the samples is statistically significant (Panel C), and these results are consistent across a range of specifications which incrementally control for additional factors. Column 2 controls for the enrollment rate of the same sex and age group in the district in 1987, and Column 3 additionally includes controls for age and its square, overall household size, the log of household consumption per capita, and the household's reported religion. Column 4 further includes the previous controls as well as the number of years the state had been ruled by the progressive party since 1992. Across all specifications, the magnitude of the difference in effects across young women and young men remains consistently around 0.016.

Using information on the other activities respondents may claim as their principal use of time available in the data, I estimate alternative outcomes for the other types of activities that school-age children could have reported as their principal activity (aside from being enrolled in school). There are several possible responses to the survey question, but they can be grouped into four types of activities: household employment, external employment, unpaid household duties/domestic work, and idle time. When estimating these outcomes for the same sample as Panel A of Table 5, we would expect negative coefficients that jointly approximate the opposite direction of the effect on enrollment to

Table 5
School enrollment rates and exposure to female leaders.

	Base (1)	+1987 enrl. rt. (2)	+ HH controls (3)	+ years prog. (4)
Panel A: Female				
Years policy exposure	0.017**	0.016**	0.020***	0.020***
	(0.007)	(0.006)	(0.007)	(0.007)
N	6332	6332	6331	6331
R^2	0.10	0.11	0.20	0.20
Mean of outcome	0.75	0.75	0.75	0.75
St. dev. of outcome	0.43	0.43	0.43	0.43
Panel B: Male				
Years policy exposure	0.003	−0.000	0.003	0.003
	(0.002)	(0.002)	(0.002)	(0.002)
N	7457	7457	7456	7456
R^2	0.07	0.08	0.14	0.14
Mean of outcome	0.85	0.85	0.85	0.85
St. dev. of outcome	0.36	0.36	0.36	0.36
Panel C: Parametric difference				
Exposure * female	0.013**	0.016***	0.017***	0.017***
	(0.006)	(0.006)	(0.006)	(0.006)
N	13,789	13,789	13,787	13,787
R^2	0.10	0.11	0.18	1.00
Mean of outcome	0.80	0.80	0.80	0.80
St. dev. of outcome	0.40	0.40	0.40	0.40

Note: Table reports coefficient estimates from equation (1) estimated via OLS. All specifications include district-pair fixed effects, an indicator for caste and its interaction with policy exposure, and an unreported constant term. Column 3 includes a vector of household level controls of linear regressors for household size, age and its square, the log of household consumption per capita, and an indicator for religion. The final specification includes a control for the number of years of state government led by the major progressive party. Standard errors are clustered by state. Significance levels are indicated by $^{*}< .1$, $^{**}< .05$, $^{***}< .01$.

indicate which activities were being displaced across the four activity groups, the main displaced activity for the now-enrolled young women was household domestic work (Column 3) (see Table 6).

I use the rich nature of the survey data to differentiate individuals by several household characteristics: household size, per-capita consumption levels, pre-policy district enrollment rates, household structure, and the highest education level among adult women in the household. Table 7 presents these estimates, showing a relative lack of heterogeneity across most of these dimensions groups – except for the strong pattern of large effects among young women in households where the highest education of adult female household members was low. This suggests the effects detected could derive from the quota policy providing specific inspiration for young women in households without educated female role models.

In Table 8, I present effects across the range of school-age children, roughly delineated into primary school (5–10 years old), secondary school (11–15 years old) and upper secondary (16–18 years old). Column 1 shows effects on the full sample, which are slightly smaller than the main estimates (replicated in Column 3) due to a roughly commensurate effect among boys in the primary-age sample, and no effect in either group among upper-secondary ages. Nevertheless, the policy appears to have had an effect across the whole school-age sample of a nearly one-percentage-point closure in the gender enrollment gap per year of exposure.

4.2. Point estimates in context: bounding the relative importance of exposure across levels

The magnitude of the results for young women of middle- and secondary-school ages is substantial: with differential effect of.013 per year and an average exposure of 10.6 years, these estimates imply that the policy was responsible for the majority of the 18 percentage-point decrease in the gender enrollment gap seen since the early 1990s. The quota policy, however, operated concurrently in multiple levels of government, and imposed rules on the composition of councils as well as council leadership seats – leaving open the question of whether the enrollment effect is attributable to the chairwomen village councils or the other female representatives in the councils either at the village or higher levels.

This question is difficult to approach empirically given the unavailability of comprehensive historical data on which villages and towns were subject to leadership reservation at different times. Nevertheless, we can compared the estimated differential enrollment effect estimated above (0.16 per an additional ten years of policy exposure) with that from Beaman et al. (2009b), who estimated a differential enrollment increase of 0.09 per ten years of exposure to village chairwomen. The estimates in this paper, however, measure the length of policy exposure, rather than exposure to female leaders *per se* (*i.e.*, the intent-to-treat effect of exposure to female politicians). To make the effects comparable, I assume that each year of policy exposure accords, on average, $\frac{1}{3}$ of a year of exposure to a female chairperson in each of the village, town, and district councils. Two comparisons are then worth discussing. I first assume that only chairwomen spur role model effects, and that enrollment effects are the same regardless of the levels of government providing exposure. Using previous estimates for magnitudes, this would imply the net effect of 10 additional years of policy exposure would similarly be 0.09 (since in expectation each year of policy exposure yields $\frac{1}{3}$ of a year of exposure to a chairwoman at each of three levels of local government). Although my estimates cannot reject this magnitude statistically, they are on the order of 50 percent larger – implying either that chairwomen in higher levels of local government generate larger role model effects, and/or that ward members in the village, town, and district bodies also generate role model effects among adolescent girls. Alternatively, we can assume that the effect comes only from exposure to village chairwomen. The estimates in Table 5 would then imply an implausibly high per-chairwoman-year effect size of 48 percentage points per additional 10 years of exposure to village chairwomen ($\frac{0.16}{\frac{1}{3}} = .48$). I thus interpret these magnitudes as evidence

Table 6
Effects of exposure to female leaders on other activities.

	Employed outside home (1)	Employed in home ent. (2)	Domestic activities (3)	All other idle time (4)
Years policy exposure	−0.000	−0.001	−0.011**	−0.004
	(0.001)	(0.001)	(0.005)	(0.003)
N	6332	6332	6332	6332
R^2	0.06	0.06	0.07	0.04
Mean of outcome	0.03	0.03	0.16	0.03
St. dev. of outcome	0.17	0.16	0.37	0.17

Note: Table reports coefficient estimates from equation (1) estimated via OLS. All specifications include district-pair fixed effects, an indicator for caste and its interaction with policy exposure, an unreported constant term and the 1987 district enrollment rate (specification in column 2 of Table 5; see notes to Table 5). Standard errors are clustered by state. Significance levels are indicated by $^{*}< .1$, $^{**}< .05$, $^{***}< .01$.

Table 7
Heterogeneous effects by household structure and characteristics.

	Low (1)	Middle (2)	High (3)
Panel A: Household size			
Years policy exposure	0.018*	0.020***	0.015***
	(0.009)	(0.006)	(0.005)
N	2175	2632	1525
R^2	0.15	0.14	0.14
Mean of outcome	0.74	0.77	0.73
St. dev. of outcome	0.44	0.42	0.44
Panel B: PPE tercile			
Years policy exposure	0.015*	0.032***	0.018**
	(0.008)	(0.007)	(0.008)
N	2042	2061	2229
R^2	0.13	0.14	0.14
Mean of outcome	0.66	0.74	0.85
St. dev. of outcome	0.47	0.44	0.36
Panel C: Tercile of 1987 enrollment rate			
Years policy exposure	0.016	0.017***	0.012*
	(0.010)	(0.003)	(0.006)
N	2088	2081	2163
R^2	0.12	0.12	0.11
Mean of outcome	0.66	0.76	0.84
St. dev. of outcome	0.48	0.43	0.37
Panel D: Ratio of boys:girls in HH			
Years policy exposure	0.011**	0.016**	0.021**
	(0.004)	(0.007)	(0.008)
N	1627	2428	2277
R^2	0.16	0.14	0.12
Mean of outcome	0.75	0.75	0.75
St. dev. of outcome	0.43	0.43	0.43
Panel E: Highest ed. of adult women			
Years policy exposure	0.021**	−0.007	−0.000
	(0.008)	(0.008)	(0.005)
N	4105	1202	1025
R^2	0.10	0.23	0.13
Mean of outcome	0.68	0.85	0.96
St. dev. of outcome	0.47	0.36	0.20

Note: Table reports coefficient estimates from equation (1) estimated via OLS. All specifications include district-pair fixed effects, an indicator for caste and its interaction with policy exposure, an unreported constant term and the 1987 district enrollment rate (specification in column 2 of Table 5; see notes to Table 5). Standard errors are clustered by state. Significance levels are indicated by $^{*}< .1$, $^{**}< .05$, $^{***}< .01$.

that there were likely substantial enrollment effects of the quota policy generated by either the ward members or chairwomen at other levels of local government. Due to historical data limitations described above, a precise assessment of the relative importance of ward member versus council leadership reservations, and exposure to female politicians across levels of government, is left for future research.

4.3. Effects on school infrastructure

A number of previous studies have shown how policies change with the application of quotas or other changes to their demographic composition. Using state-level variation in India over four decades, Pande (2003) identifies how the mandated reservations of legislative positions for minority groups increased the redistribution of resources towards these groups, demonstrating enhanced policy influence. Besley et al. (2004) found that reservation of a leadership position for SC/ST individuals increased access among SC/ST households to infrastructure and government services, while Chattopadhyay and Duflo (2004) find that the gender of local leaders affects the public goods provided in a community, and that female leaders direct public spending to infrastructure that is more relevant to their own gender.[6] Overall, the group identity of leaders has shown to matter in the type of public goods provided under the purview of the governing body, and this has been established in various contexts not limited to the Indian case (e.g., Powley, 2007; Washington, 2008).

To investigate whether the increases in young women's enrollment may be a result of increased infrastructure provision, I use school-level data from India's "District Report Cards" provided by the District Information System for Education (DISE). I use the school-level version of these data to construct measures of educational infrastructure by district. For the purpose of parsimony, I aggregate the school-level data to construct a small set of variables that allow me to test hypotheses regarding the extensive and intensive margins of schooling infrastructure. These include characteristics of schools, including the share of schools with electricity access, separate girls' lavatories, boundary walls, as well as aspects related to instruction including classroom conditions, the share of female teachers, and the student-to-teacher ratio. Finally, I estimate a measure of the extensive margin of school infrastructure, measured as the number of schools per thousand enrollees. In Panel A of Table 9, I present coefficients from the estimation of the relationship between policy exposure and the mean level of the school infrastructure measures using the specification in column 1 of Table 5. Across all measures there is limited evidence of a relationship between policy exposure and schooling infrastructure, and the majority of effects are estimated precisely enough to reject even small effect sizes.

This level effect may mask within-district heterogeneity, however, if the policy caused a reallocation of educational resources within districts. To test this, I use the detail available in the school-level DISE data to estimate the within-district *variance* of these infrastructure measures at the town and village levels to test the reallocation hypothesis. These estimates are presented in Panels B and C of Table 9, and largely rule out effects on within-district variance in schooling infrastructure. Overall, there is minimal evidence, if any, of an increase in educational infrastructure associated with longer exposure to the reform.

[6] These findings have been shown to be less robust, however, when extended to South India (Ban and Rao, 2008).

Table 8
Heterogeneous effects by age group.

	Full sample (1)	5 - 10 (2)	11 - 15 (3)	16 - 18 (4)
Panel A: Female				
Years policy exposure	0.011**	0.014***	0.016**	0.009
	(0.004)	(0.003)	(0.006)	(0.006)
N	18,778	8549	6332	3897
R^2	0.10	0.11	0.11	0.11
Mean of outcome	0.70	0.81	0.75	0.37
St. dev. of outcome	0.46	0.39	0.43	0.48
Panel B: Male				
Years policy exposure	0.005	0.011***	−0.000	0.004
	(0.003)	(0.004)	(0.002)	(0.006)
N	21,921	9745	7457	4719
R^2	0.07	0.09	0.08	0.11
Mean of outcome	0.78	0.85	0.85	0.49
St. dev. of outcome	0.42	0.36	0.36	0.50
Panel C: Parametric difference				
Exposure * female	0.007***	0.003**	0.016***	0.005
	(0.002)	(0.002)	(0.006)	(0.006)
N	40,699	18,294	13,789	8616
R^2	0.09	0.10	0.11	0.12
Mean of outcome	0.74	0.83	0.80	0.44
St. dev. of outcome	0.44	0.37	0.40	0.50

Note: Table reports coefficient estimates from equation (1) estimated via OLS. All specifications include district-pair fixed effects, an indicator for caste and its interaction with policy exposure, an unreported constant term and the 1987 district enrollment rate (specification in column 2 of Table 5; see notes to Table 5). Standard errors are clustered by state. Significance levels are indicated by *< .1, **< .05, ***< .01.

Table 9
Infrastructure analysis.

	Share with electricity (1)	Share with girls' lav. (2)	Share with boundary wall (3)	Share of classes in good condition (4)	Share of female teachers (5)	Student to teacher ratio (6)	Schools per thousand students (7)
Panel A: District means							
Years relative to first Panchayati Raj Election	−0.006	−0.021	−0.002	0.001	−0.002	−0.226	−0.009
	(0.007)	(0.013)	(0.002)	(0.007)	(0.003)	(0.319)	(0.069)
N	138	138	138	138	138	138	138
R^2	0.81	0.70	0.57	0.76	0.85	0.77	0.82
Mean of outcome	0.61	0.45	0.10	0.71	0.36	36.82	8.07
St. dev. of outcome	0.33	0.28	0.06	0.15	0.16	13.73	3.41
Panel B: SD of block means							
Years relative to first Panchayati Raj Election	0.006	0.002	0.002	−0.000	0.003	−0.091	−0.041
	(0.005)	(0.003)	(0.002)	(0.001)	(0.002)	(0.079)	(0.047)
N	138	138	138	138	138	138	138
R^2	0.63	0.65	0.64	0.63	0.44	0.66	0.58
Mean of outcome	0.09	0.10	0.05	0.08	0.09	6.63	1.67
St. dev. of outcome	0.07	0.06	0.05	0.04	0.05	4.82	1.59
Panel C: SD of village means							
Years relative to first Panchayati Raj Election	0.005	0.002	−0.003	−0.008**	−0.002	−0.163	0.036
	(0.005)	(0.006)	(0.003)	(0.003)	(0.001)	(0.418)	(0.311)
N	138	138	138	138	138	138	138
R^2	0.66	0.63	0.60	0.70	0.70	0.71	0.53
Mean of outcome	0.27	0.34	0.24	0.32	0.26	20.80	16.56
St. dev. of outcome	0.11	0.10	0.07	0.07	0.05	13.42	11.55

Note: Table reports coefficient estimates from equation TK2 estimated via OLS. All specifications include district-pair fixed effects, an indicator for caste and its interaction with policy exposure, and linear controls for household size and the log of household consumption per capita. Standard errors are clustered by state (35). Significance levels are indicated by *< .1, **< .05, ***< .01.

5. Robustness and threats to identification

I perform several tests to rule out alternative explanations for the results above as well as guard against likely threats to identification. A first-order concern – particularly when implementing a border-based identification strategy – is selective migration. I test for this in two ways. First, I use information captured in the NSS data regarding recent migration status. Since respondents provide detail on whether they have recently migrated from the same or a different state, I estimate these as outcomes in Table 10, and find no evidence of a relation-

Table 10
Relationship of policy exposure to migration status.

	Female		Male	
	Any migration in past yr. (1)	Migrated from other state (2)	Any migration in past yr. (3)	Migrated from other state (4)
Years policy exposure	−0.00028	0.00053	0.00001	−0.00002
	(0.00049)	(0.00038)	(0.00066)	(0.00008)
N	6332	6332	7457	7457
R^2	0.03	0.04	0.04	0.03
Mean of outcome	0.007	0.002	0.011	0.001
St. dev. of outcome	0.084	0.043	0.105	0.031

Note: Table reports coefficient estimates from equation (1) estimated via OLS. All specifications include district-pair fixed effects, an indicator for caste and its interaction with policy exposure, an unreported constant term and the 1987 district enrollment rate (specification in column 2 of Table 5; see notes to Table 5). Standard errors are clustered by state. Significance levels are indicated by $^{*}< .1$, $^{**}< .05$, $^{***}< .01$.

Table 11
Adult education levels and policy exposure.

	Base (1)	+1987 enrl. rt. (2)	+ HH controls (3)	+ years prog. (4)
Panel A: Highest ranking woman's education level				
Years policy exposure	0.029	0.022	0.037	0.041
	(0.020)	(0.020)	(0.027)	(0.028)
N	6021	6021	6020	6020
R^2	0.17	0.17	0.24	0.24
Mean of outcome	1.62	1.62	1.62	1.62
St. dev. of outcome	2.99	2.99	2.99	2.99
Panel B: Head of household education level				
Years policy exposure	0.030	0.029	0.056	0.052
	(0.032)	(0.031)	(0.041)	(0.040)
N	6331	6331	6330	6330
R^2	0.10	0.10	0.16	0.16
Mean of outcome	3.53	3.53	3.53	3.53
St. dev. of outcome	4.13	4.13	4.13	4.13

Note: Table reports coefficient estimates from equation (1) estimated via OLS. All specifications include district-pair fixed effects, an indicator for caste and its interaction with policy exposure, and linear controls for household size and the log of household consumption per capita. Standard errors are clustered by state. Significance levels are indicated by $^{*}< .1$, $^{**}< .05$, $^{***}< .01$.

Table 12
Falsification: educational attainment of cohorts out of school before policy start.

	Female		Male	
	30–34 yrs. old (1)	35–40 yrs. old (2)	30–34 yrs. old (3)	35–40 yrs. old (4)
Years policy exposure	0.040	0.002	0.059	0.025
	(0.035)	(0.040)	(0.052)	(0.034)
N	4518	4331	4386	3919
R^2	0.21	0.22	0.14	0.15
Mean of outcome	2.99	2.24	5.74	4.84
St. dev. of outcome	3.95	3.56	4.49	4.38

Note: Table reports coefficient estimates from equation (1) estimated via OLS. All specifications include district-pair fixed effects, an indicator for caste and its interaction with policy exposure, an unreported constant term and the 1987 district enrollment rate (specification in column 2 of Table 5; see notes to Table 5). Standard errors are clustered by state. Significance levels are indicated by $^{*}< .1$, $^{**}< .05$, $^{***}< .01$.

ship between policy exposure and recent household migration behavior. Second, in Table 11, I exploit the household structure information in the survey data to estimate whether the policy exposure has any relationship to the years of education of the household's highest-ranking woman and the household head, in Panels A and B, respectively. If selective migration played a role in generating the main results, we might expect that these two measures would be correlated with the ability or desire to move to more female(-education) friendly states. Nevertheless, I find no evidence of a relationship between the policy and the education level of the adults of the cohorts of children who exhibited an enrollment response to policy exposure. I also falsify the hypothesis that policy exposure was related to educational attainment of older cohorts of women who had already completed their education by the time of its first introduction by estimating the educational attainment directly among the sample of adults who were between 30 and 40 years old in 2007 (shown in Table 12). As expected, we see no mean-

ingful effect on cohorts who had largely already completed schooling by the time of the first state's introduction of the quota policy. Finally, I find no meaningful difference when fully absorbing effects among the sample of children in West Bengal, which was the earliest state to implement the quota regime – concurrently with the passing of the national law – to dispel the concern that results are driven by this state (see Appendix Table 2).

6. Conclusion

This paper investigates the relationship between quotas for women in three tiers of local government and their effects on young women's educational enrollment. I find that India's flagship policy mandating representation quotas for women in local government increased school enrollment rates of adolescent girls substantially, with no commensurate increase in enrollment of adolescent boys. I find no evidence of effects among schooling infrastructure, the labor market, or indicators of women's status in the household. These increases in enrollment are seen alongside reductions in idle time and non-labor force household duties, and the effects were strongest among poor families and those with uneducated adult women. There is no evidence of changes in school infrastructure either overall or in terms of a reallocation of resources within districts.

The focal policy implemented a one-third quota for women among governance bodies (ward members) as well as for each constituency's single-seat leadership position, and this quota rule was applied across three levels of local government at the village, town, and district levels. This paper estimates the combined effect of exposure to female ward members and leaders across all three levels of local government. I am able to compare effect magnitudes with previously-estimated average treatment effects of exposure to a female council leader at the lowest (village) level of local governance, and the estimated magnitudes imply that a substantial portion of the net effect of exposure to female politicians on girls' school enrollment comes from female leaders serving in reserved positions other than the village chairperson seat. Future work should investigate whether this additional enrollment effect is causally linked to exposure to village ward members, or politicians in higher levels of government.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jdeveco.2018.08.004.

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